

Mathematical, Physical, and Control Documentation

Advanced Simulation Environment for Control Systems

Simulation Model Technical Documentation

September 6, 2026

Contents

Introduction and Purpose of the Documentation	2
1 Gantry Crane (Crane Anti-Sway Control)	2
1.1 Physical Description of the System	2
1.2 Full Derivation of the Mathematical Model	2
1.2.1 Simplified Form in Simulation Implementation	4
1.3 Physical Parameters of the Model	4
1.4 Dual-PD Anti-Sway Controller: Operating Principle and Justification . . .	4
2 Furuta Pendulum (Rotary Inverted Pendulum - State-Feedback Control)	5
2.1 Physical Description of the System	5
2.2 Full Derivation of the Mathematical Model via Lagrange's Method	5
2.3 Physical Parameters of the Model	7
2.4 Linear State-Feedback Controller: Operating Principle and Justification . .	7
3 Quadcopter (Cascade PID Position Control)	8
3.1 Physical Description of the System	8
3.2 Full Derivation of Quadcopter Dynamics and Cascade Control	8
3.3 Physical Parameters of the Model	9
3.4 Cascade SafePID Controller with Derivative on Measurement	9
4 Ball and Plate (2D Ball Balancing PD)	9
4.1 Physical Description of the System	9
4.2 Full Derivation of the Rolling Ball Model	9
4.3 Physical Parameters of the Model	10
4.4 Two-Axis PD Controller with Median Filtering: Justification	10
5 Satellite with Reaction Wheel (Reaction Wheel Attitude Control)	11
5.1 Physical Description of the System	11
5.2 Full Derivation of Momentum Exchange Dynamics	11
5.3 Physical Parameters of the Model	11
5.4 PID Controller with Gyroscope Angular Rate Feedback	12

Introduction and Purpose of the Documentation

This document constitutes a compendium of knowledge regarding the physical dynamics models and control algorithms implemented in the simulation environment. The primary objective of this documentation is to provide rigorous formal derivations of the differential equations describing the dynamic behavior of the test plants, along with a theoretical analysis of the control structures applied.

Five representative dynamic systems of varying complexity are presented (ranging from single- to multi-variable, non-linear, and open-loop unstable systems):

1. **Gantry Crane (Crane Anti-Sway Control):** A system with constraints and an unstable subspace (payload sway).
2. **Furuta Pendulum (Rotary Inverted Pendulum):** A highly non-linear, coupled system with strong centrifugal and Coriolis dynamics.
3. **Planar Quadcopter:** A cascade control system regulating position through spatial attitude.
4. **Ball and Plate (2D Ball Balancing):** A two-axis unstable, non-stationary system with a point-contact rolling interface.
5. **Satellite with Reaction Wheel:** An attitude control system based on angular momentum exchange and reaction torque.

1 Gantry Crane (Crane Anti-Sway Control)

1.1 Physical Description of the System

The gantry crane system consists of a motor-driven cart of mass M (`m_cart`), moving along a horizontal rail along the X -axis, and a payload treated as a point mass m (`m_load`), suspended on a massless, rigid cable of length L (`length`). The position of the cart is defined by the coordinate x , while the angular deflection of the cable from the vertical is described by the angle θ .

1.2 Full Derivation of the Mathematical Model

The Cartesian coordinates of the suspended payload (x_m, y_m) in the reference frame attached to the rail are expressed as:

$$x_m = x + L \sin \theta \quad (1)$$

$$y_m = -L \cos \theta \quad (2)$$

Differentiating the above relationships with respect to time t , we obtain the payload velocities in Cartesian coordinates:

$$\dot{x}_m = \dot{x} + L\dot{\theta} \cos \theta \quad (3)$$

$$\dot{y}_m = L\dot{\theta} \sin \theta \quad (4)$$

The square of the total payload velocity is:

$$v_m^2 = \dot{x}_m^2 + \dot{y}_m^2 = (\dot{x} + L\dot{\theta} \cos \theta)^2 + (L\dot{\theta} \sin \theta)^2 = \dot{x}^2 + 2L\dot{x}\dot{\theta} \cos \theta + L^2\dot{\theta}^2 \quad (5)$$

The total kinetic energy of the system T is the sum of the kinetic energies of the cart and the payload:

$$T = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}mv_m^2 = \frac{1}{2}(M + m)\dot{x}^2 + mL\dot{x}\dot{\theta} \cos \theta + \frac{1}{2}mL^2\dot{\theta}^2 \quad (6)$$

The potential energy of the system V (assuming the reference level $y = 0$ at the rail height) is:

$$V = -mgL \cos \theta \quad (7)$$

The Lagrangian function $\mathcal{L} = T - V$ takes the form:

$$\mathcal{L}(x, \theta, \dot{x}, \dot{\theta}) = \frac{1}{2}(M + m)\dot{x}^2 + mL\dot{x}\dot{\theta} \cos \theta + \frac{1}{2}mL^2\dot{\theta}^2 + mgL \cos \theta \quad (8)$$

We apply the Euler-Lagrange equations of the second kind for the generalized coordinates $q_1 = x$ and $q_2 = \theta$:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i \quad (9)$$

For coordinate x (where the generalized force $Q_x = F$):

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = (M + m)\dot{x} + mL\dot{\theta} \cos \theta \quad (10)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) = (M + m)\ddot{x} + mL\ddot{\theta} \cos \theta - mL\dot{\theta}^2 \sin \theta \quad (11)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad (12)$$

Which yields the first equation of motion:

$$(M + m)\ddot{x} + mL\ddot{\theta} \cos \theta - mL\dot{\theta}^2 \sin \theta = F \quad (13)$$

For coordinate θ (where the generalized force $Q_\theta = 0$):

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mL\dot{x} \cos \theta + mL^2\dot{\theta} \quad (14)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = mL\ddot{x} \cos \theta - mL\dot{x}\dot{\theta} \sin \theta + mL^2\ddot{\theta} \quad (15)$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -mL\dot{x}\dot{\theta} \sin \theta - mgL \sin \theta \quad (16)$$

We obtain the second equation of motion:

$$mL^2\ddot{\theta} + mL\ddot{x} \cos \theta + mgL \sin \theta = 0 \implies L\ddot{\theta} + \ddot{x} \cos \theta + g \sin \theta = 0 \quad (17)$$

1.2.1 Simplified Form in Simulation Implementation

In the simulation engine, an assumption of small angular velocities ($\dot{\theta}^2 \approx 0$) is adopted, along with an approximation of the direct impact of the force F on cart acceleration $\ddot{x} \approx \frac{F}{M+m}$. Substituting this into equation (17), the angular acceleration $\alpha = \ddot{\theta}$ is computed:

$$\alpha = \frac{-g \sin \theta - \frac{F}{M+m} \cos \theta}{L} \quad (18)$$

The calculated angular acceleration is substituted back into the full cart equation (13) to yield the linear acceleration $a_x = \ddot{x}$:

$$a_x = \frac{F + mL\alpha \cos \theta}{M + m} \quad (19)$$

Discrete state integration is carried out via Euler's method with viscous velocity damping (damping coefficients 0.98 and 0.985):

$$v_x(t + dt) = 0.98 \cdot (v_x(t) + a_x \cdot dt) \quad (20)$$

$$\omega(t + dt) = 0.985 \cdot (\omega(t) + \alpha \cdot dt) \quad (21)$$

The boundary condition for travel limits $|x| > x_{limit}$ enforces position relaxation and an elastoplastic rebound: $v_x \leftarrow -0.2 \cdot v_x$.

1.3 Physical Parameters of the Model

Table 1: Physical parameters and variables of the Crane model

Symbol	Parameter Description	Code Variable	Value	Unit
M	Cart mass	<code>m_cart</code>	2.0	kg
m	Payload mass	<code>m_load</code>	0.5	kg
L	Cable length	<code>length</code>	1.0	m
g	Gravitational acceleration	<code>g</code>	9.81	m/s ²
x_{limit}	Track movement range limit	<code>x_limit</code>	2.5	m
F_{max}	Driving force limit	<code>max_force</code>	30.0	N

1.4 Dual-PD Anti-Sway Controller: Operating Principle and Justification

The gantry crane is an underactuated system—it possesses two degrees of freedom (x, θ), but only a single actuator generating force F . A conventional position controller applied solely to the cart would induce prolonged, undamped payload oscillations.

A Dual-PD controller architecture is employed, combining a positioning loop with a sway damping loop:

$$F_{pos} = K_{p,pos} \cdot (x_{set} - x) - K_{d,pos} \cdot v_x \quad (22)$$

$$F_{angle} = K_{p,angle} \cdot \theta + K_{d,angle} \cdot \omega \quad (23)$$

$$F = \text{sat}_{[-F_{max}, F_{max}]}(F_{pos} + F_{angle}) \quad (24)$$

Justification for Structure Selection:

1. **No need for an integral term (I):** An integral term in the presence of force limits (F_{max}) would generate windup, which would degrade the load damping phase. Static friction in the model is negligible, so the P term is sufficient to bring the tracking error to zero.
2. **Positive angle feedback (F_{angle}):** Moving the cart in the direction of the load deflection generates the desired centrifugal and gravitational acceleration, transferring energy from the pendulum module to the cart's momentum, which effectively and quickly suppresses oscillations ($\theta \rightarrow 0$).

2 Furuta Pendulum (Rotary Inverted Pendulum - State-Feedback Control)

2.1 Physical Description of the System

The Furuta pendulum (rotary inverted pendulum) consists of a motor-driven horizontal arm of total length L_r , mass m_r , and moment of inertia J_r relative to its axis of rotation, rotating in the horizontal plane by an angle θ_1 (actuated by motor torque u). Attached via a joint to the tip of this arm is a second link—a pendulum of total length L_p , mass m_p , and moment of inertia J_p relative to its center of mass—which rotates freely in a vertical plane (angle θ_2).

The upright vertical position corresponds to $\theta_2 = 0$, while the stable downward position corresponds to $\theta_2 = \pi$.

2.2 Full Derivation of the Mathematical Model via Lagrange's Method

To derive the complete equations of motion, we define the Cartesian position vectors of the centers of mass for both the driven rotary arm ($\mathbf{r}_r$) and the pendulum link ($\mathbf{r}_p$) in a fixed world reference frame:

1. **Center of Mass of the Rotary Arm (G_r):**

$$\mathbf{r}_r = \begin{bmatrix} \frac{1}{2}L_r \cos \theta_1 \\ \frac{1}{2}L_r \sin \theta_1 \\ 0 \end{bmatrix} \quad (25)$$

2. **Center of Mass of the Pendulum Link (G_p):**

$$\mathbf{r}_p = \begin{bmatrix} L_r \cos \theta_1 + \frac{1}{2}L_p \sin \theta_2 \sin \theta_1 \\ L_r \sin \theta_1 - \frac{1}{2}L_p \sin \theta_2 \cos \theta_1 \\ \frac{1}{2}L_p \cos \theta_2 \end{bmatrix} \quad (26)$$

Taking time derivatives of $\mathbf{r}_r$ and $\mathbf{r}_p$, we compute the respective translational velocity vectors $\mathbf{v}_r = \dot{\mathbf{r}}_r$ and $\mathbf{v}_p = \dot{\mathbf{r}}_p$:

$$v_r^2 = \|\dot{\mathbf{r}}_r\|^2 = \frac{1}{4}L_r^2\dot{\theta}_1^2 \quad (27)$$

$$v_p^2 = \|\dot{\mathbf{r}}_p\|^2 = \left(L_r^2 + \frac{1}{4}L_p^2 \sin^2 \theta_2 \right) \dot{\theta}_1^2 + \frac{1}{4}L_p^2 \dot{\theta}_2^2 - L_r L_p \cos \theta_2 \dot{\theta}_1 \dot{\theta}_2 \quad (28)$$

The total kinetic energy $T = T_{\text{trans}} + T_{\text{rot}}$ of the system combines translational motion of the centers of mass and rotational movement about the axes:

$$T = \frac{1}{2}J_r \dot{\theta}_1^2 + \frac{1}{2}m_p v_p^2 + \frac{1}{2}J_p \dot{\theta}_2^2 \quad (29)$$

Substituting v_p^2 yields the expanded kinetic energy expression:

$$T = \frac{1}{2} \left(J_r + m_p L_r^2 + \frac{1}{4}m_p L_p^2 \sin^2 \theta_2 \right) \dot{\theta}_1^2 + \frac{1}{2} \left(J_p + \frac{1}{4}m_p L_p^2 \right) \dot{\theta}_2^2 - \frac{1}{2}m_p L_r L_p \cos \theta_2 \dot{\theta}_1 \dot{\theta}_2 \quad (30)$$

The potential energy V , measured relative to the horizontal plane containing the rotary arm joint ($z = 0$), depends strictly on the height of the pendulum's center of mass:

$$V = \frac{1}{2}m_p g L_p \cos \theta_2 \quad (31)$$

The Lagrangian function $\mathcal{L} = T - V$ is formulated in compact matrix notation as:

$$\mathcal{L}(\theta, \dot{\theta}) = \frac{1}{2} \begin{bmatrix} \dot{\theta}_1 & \dot{\theta}_2 \end{bmatrix} \begin{bmatrix} d_{11}(\theta_2) & d_{12}(\theta_2) \\ d_{21}(\theta_2) & d_{22} \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} - \frac{1}{2}m_p g L_p \cos \theta_2 \quad (32)$$

where the elements of the symmetric mass-inertia matrix $D(\theta_2)$ are:

$$d_{11}(\theta_2) = J_r + m_p L_r^2 + \frac{1}{4}m_p L_p^2 \sin^2 \theta_2 \quad (33)$$

$$d_{12}(\theta_2) = d_{21}(\theta_2) = -\frac{1}{2}m_p L_r L_p \cos \theta_2 \quad (34)$$

$$d_{22} = J_p + \frac{1}{4}m_p L_p^2 \quad (35)$$

Applying the Euler-Lagrange equations of the second kind with non-conservative forces (actuation torque u and joint viscous damping coefficients b_r, b_p):

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i} \right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = Q_i \quad (36)$$

This yields the system dynamics in standard matrix form:

$$\begin{bmatrix} d_{11}(\theta_2) & d_{12}(\theta_2) \\ d_{21}(\theta_2) & d_{22} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} + \begin{bmatrix} c_1(\theta_2, \omega_1, \omega_2) \\ c_2(\theta_2, \omega_1) \end{bmatrix} + \begin{bmatrix} 0 \\ g_2(\theta_2) \end{bmatrix} = \begin{bmatrix} u - b_r \omega_1 \\ -b_p \omega_2 \end{bmatrix} \quad (37)$$

where $\omega_1 = \dot{\theta}_1$, $\omega_2 = \dot{\theta}_2$, $\alpha_1 = \ddot{\theta}_1$, $\alpha_2 = \ddot{\theta}_2$, and:

$$c_1(\theta_2, \omega_1, \omega_2) = \frac{1}{2}m_p L_p^2 \sin(\theta_2) \cos(\theta_2) \omega_1 \omega_2 + \frac{1}{2}m_p L_r L_p \sin(\theta_2) \omega_2^2 \quad (38)$$

$$c_2(\theta_2, \omega_1) = -\frac{1}{4}m_p L_p^2 \sin(\theta_2) \cos(\theta_2) \omega_1^2 \quad (39)$$

$$g_2(\theta_2) = -\frac{1}{2}m_p g L_p \sin(\theta_2) \quad (40)$$

Rearranging into generalized effective force terms f_1 and f_2 :

$$f_1 = u - b_r \omega_1 - c_1(\theta_2, \omega_1, \omega_2) \quad (41)$$

$$f_2 = -b_p \omega_2 - c_2(\theta_2, \omega_1) - g_2(\theta_2) \quad (42)$$

The angular accelerations $\alpha = [\alpha_1, \alpha_2]^T$ are solved explicitly by inverting $D(\theta_2)$:

$$\det D(\theta_2) = d_{11}(\theta_2)d_{22} - d_{12}^2(\theta_2) \quad (43)$$

$$\alpha_1 = \frac{d_{22}f_1 - d_{12}(\theta_2)f_2}{\det D(\theta_2)}, \quad \alpha_2 = \frac{-d_{21}(\theta_2)f_1 + d_{11}(\theta_2)f_2}{\det D(\theta_2)} \quad (44)$$

To avoid numerical instabilities caused by stiff differential dynamics in simulation, an explicit Euler sub-stepping integration scheme is implemented using a sub-step time interval $dt_{\text{sub}} = 0.0005$ s.

2.3 Physical Parameters of the Model

Table 2: Physical parameters of the Furuta Pendulum model

Symbol	Parameter Description	Code Variable	Value	Unit
L_r	Rotary arm length	L_r	0.25	m
m_r	Rotary arm mass	m_r	0.20	kg
J_r	Rotary arm moment of inertia	J_r	0.005	kg·m ²
L_p	Pendulum link length	L_p	0.30	m
m_p	Pendulum mass	m_p	0.12	kg
J_p	Pendulum moment of inertia (CoM)	J_p	0.0015	kg·m ²
b_r	Rotary joint friction coefficient	b_r	0.001	N·m·s/rad
b_p	Pendulum joint friction coefficient	b_p	0.0005	N·m·s/rad
θ_{act}	Controller activation angle limit	activation_angle	20°	deg

2.4 Linear State-Feedback Controller: Operating Principle and Justification

Near the upright unstable equilibrium point ($\mathbf{x}_0 = [\theta_{1,\text{set}}, 0, 0, 0]^T$), the system exhibits strong non-linear cross-couplings across its four state variables:

$$\mathbf{x} = \begin{bmatrix} e_{\theta_1} \\ \omega_1 \\ \theta_2 \\ \omega_2 \end{bmatrix} = \begin{bmatrix} \text{normalize_angle}(\theta_1 - \theta_{1,\text{set}}) \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \end{bmatrix} \quad (45)$$

Linearizing the nonlinear dynamic equations around $\theta_2 = 0, \omega_1 = 0, \omega_2 = 0$ yields a continuous-time state-space model $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$. A state-feedback gain vector $\mathbf{K} = [K_0, K_1, K_2, K_3]$ is synthesized (e.g., via pole placement or optimal gain selection) to place the closed-loop poles $\det(s\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K})) = 0$ in the open left-half complex plane.

The linear state-feedback control law computing the motor torque demand u is defined as:

$$u = -\mathbf{K}\mathbf{x} = -(K_0 \cdot \text{normalize_angle}(\theta_1 - \theta_{1,\text{set}}) + K_1 \cdot \omega_1 + K_2 \cdot \theta_2 + K_3 \cdot \omega_2) \quad (46)$$

Justification for State-Feedback Control and Switching Threshold:

1. **MIMO Coupling and Underactuation:** Independent single-input single-output (SISO) loops fail to stabilize the upright pose, because horizontal rotations of the arm induce non-linear centrifugal and Coriolis reaction forces directly affecting pendulum dynamics. Full state feedback actively decouples and dampens these dynamic cross-terms.
2. **Angle Normalization and Activation Window** ($|\theta_2| \leq \theta_{\text{act}}$): The linearized feedback gain vector $\mathbf{K}$ is calculated locally for small perturbations around $\theta_2 = 0$. In the downward region ($|\theta_2| > 20^\circ$), linear feedback generates control signals of incorrect sign, driving the system into instability. To ensure safety, the controller disengages ($u = 0$) whenever $|\theta_2| > \theta_{\text{act}}$.

3 Quadrocopter (Cascade PID Position Control)

3.1 Physical Description of the System

The model describes a quadrocopter constrained to a 2D vertical plane. The system lacks direct horizontal propulsion—total thrust produced by the rotors is always directed along the frame’s vertical body axis. Motion along the X and Y axes is actuated via hull inclination (angles ϕ – roll and θ – pitch).

3.2 Full Derivation of Quadrocopter Dynamics and Cascade Control

Total thrust force T_{total} balances gravity mg in hover. Upon tilting the drone by roll angle ϕ and pitch angle θ , horizontal thrust force components driving lateral acceleration are:

$$F_x = T_{\text{total}} \sin \phi \cos \theta \approx mg \tan \phi \quad (47)$$

$$F_y = -T_{\text{total}} \sin \theta \approx -mg \tan \theta \quad (48)$$

Dynamic equations of motion incorporating linear air drag with coefficient $b = 3.0 \text{ s}^{-1}$:

$$a_x = \ddot{x} = g \tan \phi - 3.0 \cdot v_x \quad (49)$$

$$a_y = \ddot{y} = -g \tan \theta - 3.0 \cdot v_y \quad (50)$$

Internal powertrain dynamics (motors + electronic speed controllers) are modeled as a 1st order low-pass filter delaying angular response with time constant $\tau = 0.04 \text{ s}$:

$$\phi(t + dt) = \phi(t) + (\phi_{\text{set}} - \phi(t)) \cdot \frac{dt}{\tau + dt} \quad (51)$$

$$\theta(t + dt) = \theta(t) + (\theta_{\text{set}} - \theta(t)) \cdot \frac{dt}{\tau + dt} \quad (52)$$

Integration of velocity and position is handled via Euler’s method:

$$v_x(t + dt) = v_x(t) + a_x dt, \quad x(t + dt) = x(t) + v_x dt \quad (53)$$

Table 3: Physical parameters of the Quadrocopter model

Symbol	Parameter Description	Code Variable	Value	Unit
L	Drone arm length	<code>arm_len</code>	0.25	m
g	Gravitational acceleration	<code>g</code>	9.81	m/s ²
τ	Gyro/actuator response time constant	<code>tau</code>	0.04	s
b	Air drag coefficient	<code>drag</code>	3.0	1/s
limit	Tilt angle saturation limit	<code>pid_limit</code>	15°	deg

3.3 Physical Parameters of the Model

3.4 Cascade SafePID Controller with Derivative on Measurement

The outer loop regulates position (x, y) , outputting target setpoint attitude angles $(\phi_{set}, \theta_{set})$ that serve as reference inputs to the inner loop.

The implemented **SafePID** algorithm incorporates a *Derivative on Measurement* modification:

$$e_x(t) = x_{set} - x(t) \quad (54)$$

$$I_x(t) = \text{sat}_{[-5,5]}(I_x(t-1) + e_x(t) \cdot dt) \quad (\text{Anti-Windup}) \quad (55)$$

$$D_x(t) = \frac{x(t) - x(t-dt)}{dt} \quad (56)$$

$$\phi_{set} = \text{sat}_{[-15^\circ, 15^\circ]}(K_p e_x(t) + K_i I_x(t) - K_d D_x(t)) \quad (57)$$

Justification for Applied Safeguards:

1. **Derivative on Measurement Modification:** Standard error differentiation $\frac{de(t)}{dt}$ produces a sharp derivative spike (known as *derivative kick*) upon step changes in setpoint x_{set} , triggering abrupt tilts and potential instability. Differentiating state measurement $x(t)$ directly eliminates this effect.
2. **Integral Anti-Windup:** Restricting integral accumulation to the tight range $[-5.0, 5.0]$ prevents actuator saturation while operating far from the setpoint.

4 Ball and Plate (2D Ball Balancing PD)

4.1 Physical Description of the System

The plant consists of a square plate of side length L , tilted along two axes by angles θ_{pitch} (around Y-axis, driving motion along X) and θ_{roll} (around X-axis, driving motion along Y). A solid sphere of mass m and radius r rolls across the plate without slipping.

4.2 Full Derivation of the Rolling Ball Model

For a solid sphere of radius r and mass m , its moment of inertia is $J = \frac{2}{5}mr^2$.

Applying energy conservation or Newton-Euler rotational-translational dynamic equations for non-slip rolling ($a = \alpha \cdot r$):

$$ma_x = F_g - F_{fric} \implies m\ddot{x} = mg \sin \theta_{pitch} - \frac{J\ddot{x}}{r^2} \quad (58)$$

$$\left(m + \frac{J}{r^2}\right) \ddot{x} = mg \sin \theta_{pitch} \implies \left(m + \frac{2}{5}m\right) \ddot{x} = mg \sin \theta_{pitch} \quad (59)$$

$$\frac{7}{5}m\ddot{x} = mg \sin \theta_{pitch} \implies \ddot{x} = \frac{5}{7}g \sin \theta_{pitch} \quad (60)$$

Accounting for viscous rolling friction with coefficient $b = 1.2 \text{ s}^{-1}$, state space dynamic equations become:

$$\ddot{x} = -\frac{5}{7}g \sin(\theta_{pitch}) - b\dot{x} \quad (61)$$

$$\ddot{y} = \frac{5}{7}g \sin(\theta_{roll}) - b\dot{y} \quad (62)$$

Boundary collision conditions incorporate sphere geometry: $|x| \geq \frac{L}{2} - r$. Upon impact, elastic deformation occurs governed by restitution coefficient $e = 0.4$:

$$v_{post} = -e \cdot v_{pre} \quad (63)$$

4.3 Physical Parameters of the Model

Table 4: Parameters of the Ball and Plate model

Symbol	Parameter Description	Code Variable	Value	Unit
L	Plate side length	<code>plate_size</code>	1.4	m
r	Ball radius	<code>ball_radius</code>	0.09	m
g	Gravitational acceleration	<code>g</code>	9.81	m/s ²
b	Rolling resistance coefficient	<code>damping</code>	1.2	1/s
e	Boundary restitution coefficient	<code>restitution</code>	0.4	-

4.4 Two-Axis PD Controller with Median Filtering: Justification

The system is governed by independent PD controllers along each axis, setting target plate angles:

$$\theta_{pitch}(t) = -\text{sat}_{[-18^\circ, 18^\circ]}(K_p(x_{set} - x) - K_d v_x) \quad (64)$$

$$\theta_{roll}(t) = \text{sat}_{[-18^\circ, 18^\circ]}(K_p(y_{set} - y) + K_d v_y) \quad (65)$$

Justification for Structure Selection:

1. **Omission of Integral Action (I):** Ball movement on an inclined plane represents a double integrator plant ($a \rightarrow v \rightarrow x$). Adding a third integral term in the controller introduces a -270° phase shift, inducing severe oscillations and instability near the setpoint.
2. **Median Filter (Window $N = 5$):** Input signals from physical IMU sensors contain impulse noise. A median filter effectively eliminates peak noise artifacts without introducing phase delays characteristic of low-pass filters.

5 Satellite with Reaction Wheel (Reaction Wheel Attitude Control)

5.1 Physical Description of the System

A satellite with moment of inertia I_{sat} controls its angular orientation θ_{sat} by adjusting the angular velocity of an internal flywheel with moment of inertia I_{wheel} . In accordance with the conservation of angular momentum, motor electromagnetic torque M_{motor} accelerates the wheel, inducing an equal and opposite reaction torque on the satellite bus.

5.2 Full Derivation of Momentum Exchange Dynamics

Total angular momentum of the closed system (excluding external gravitational or aerodynamic torques) is:

$$H_{total} = I_{sat}\omega_{sat} + I_{wheel}\omega_{wheel_abs} = const \quad (66)$$

where the absolute angular velocity of the flywheel in space is $\omega_{wheel_abs} = \omega_{sat} + \omega_{wheel}$.

Differentiating with respect to time t in the presence of internal motor torque M_{motor} and bearing friction $M_f = b \cdot \omega_{wheel}$:

$$I_{sat}\alpha_{sat} = M_{motor} - b \cdot \omega_{wheel} \implies \alpha_{sat} = \frac{M_{motor} - b \cdot \omega_{wheel}}{I_{sat}} \quad (67)$$

For the reaction wheel, relative angular acceleration $\alpha_{wheel_rel} = \dot{\omega}_{wheel}$ is given by:

$$\alpha_{wheel_rel} = -\frac{M_{motor}}{I_{wheel}} - \alpha_{sat} + \frac{b \cdot \omega_{wheel}}{I_{wheel}} \quad (68)$$

Wheel velocity is physically constrained by motor/inverter saturation $\omega_{max_wheel} = 300$ rad/s. Upon reaching this threshold, torque M_{motor} is clamped to prevent further wheel acceleration.

5.3 Physical Parameters of the Model

Table 5: Physical parameters of the Satellite model

Symbol	Parameter Description	Code Variable	Value	Unit
I_{sat}	Satellite moment of inertia	I_sat	2.5	kg·m ²
I_{wheel}	Flywheel moment of inertia	I_wheel	0.3	kg·m ²
ω_{max}	Max wheel speed limit	max_wheel_speed	300.0	rad/s
b	Bearing friction coefficient	bearing_friction	0.01	N·m·s/rad
M_{max}	Max motor torque limit	max_torque	25.0	N·m

5.4 PID Controller with Gyroscope Angular Rate Feedback

Attitude control is executed via a dedicated PID controller operating on angular error mapped onto a circle:

$$e_\theta(t) = \text{angle_difference}(\theta_{set}, \theta_{sat}) = (\theta_{set} - \theta_{sat} + \pi) \pmod{2\pi} - \pi \quad (69)$$

$$I(t) = \text{sat}_{[-5,5]}(I(t-1) + e_\theta(t) \cdot dt) \quad (70)$$

$$M_{motor} = \text{sat}_{[-M_{max}, M_{max}]}(K_p e_\theta(t) + K_i I(t) - K_d \omega_{sat}(t)) \quad (71)$$

Justification for Design Solution:

1. **Angular Error Normalization (angle_difference):** Prevents the satellite from attempting a 350° rotation when a short -10° maneuver across the $\pm\pi$ boundary suffices.
2. **Direct ω_{sat} Feedback over Error Differentiation:** Basing the derivative path directly on gyroscope-measured satellite velocity ω_{sat} provides pure viscous damping (rate damping), raising system stability margin without sensitivity to setpoint step changes.